MTH220 2022 S2 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

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* ICT231 Data Structures and Algorithms I
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* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH220

Timed Online Assignment - July Semester 2022

Statistical Methods and Inference

Thursday, 24 November 2022 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **NINE (9)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and Not Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

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7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

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Answer all questions. (Total 100 marks)

Question 1a

The sales supervisor of a popular western food restaurant is interested to find out if there are ways to increase the sales of the spring chicken. He hypothesizes that customers prefer bigger chickens and thus more likely to order the spring chicken dish. Over the course of 13 weeks, he calculates the Average Weight of all the chicken sold for that week and compares it against the Weekly Sales amount for spring chickens for that week with data shown in Table Q1(a).

Week	**Average Chicken Weight, X (kg)**	**Weekly Sales, Y ($)**
1	1.21	100
2	1.21	120
3	1.83	160
4	1.79	130
5	1.83	120
6	1.32	170
7	2.64	180
8	2.32	200
9	2.39	210
10	2.17	190
11	2.12	220
12	2.86	225
13	2.97	240

Table Q1(a)

Question 1a(i)

Construct a scatter plot and comment on the results.

(3 marks)

Question 1a(ii)

Assuming a linear relationship, use the least-squares method to determine the regression coefficients and the linear regression model.

(5 marks)

Question 1a(iii)

Predict the weekly sales of spring chicken if the chicken has a weight of 1.5kg.

(1 mark)

Question 1a(iv)

Compute the Pearson Correlation Coefficient between X and Y. Comment on goodness of fit of the model.

(3 marks)

Question 1b

A 95% confidence interval constructed for the mean mass of the pastry from a random sample of 75 pastries from a bakery is (91.9631, 96.0369).

Question 1b(i)

Calculate the sample mean and sample variance of the mass of pastries.

(3 marks)

Question 1b(ii)

The pastries were stated to be 90g on the packaging. The director felt that the mass was underestimated and tasked the production manager (you) to investigate further. State appropriate null and alternative hypothesis for this investigation. By using the data from the existing sample of 75 pastries, determine, at the 10% level of significance, whether the company has understated the mean mass of the pastries.

(6 marks)

Question 1b(iii)

To increase the sales of the pastry, the company revises the mean mass of the pastries printed on the packets to q grams, where q is as positive integer. A manufacturing manager from the company wishes to test whether the mean mass of the pastries is q grams.

Using data from the existing sample of 75 pastries, find the smallest value of q (where q is a integer) so that there is no reason for the officer to reject the revised printed mass at the 1% level of significance.

(4 marks)

Question 2a

A secondary school teacher was interested to find out about the proficiency of his math students in Class A and B. He gathered results from past math tests to evaluate both classes. The objective is to determine whether both classes are on average equally proficient.

A random sample of 11 tests were taken from Class A, resulting in mean score of 80 with standard deviation of 3.

A random sample of 14 tests were taken from Class B, resulting in mean score of 78 with standard deviation of 4.

Assume that the population test scores from the two classes are approximately normal with same population variance.

Question 2a(i)

Formulate the null and alternative hypothesis.

(2 marks)

Question 2a(ii)

Compute the pooled variance.

(2 marks)

Question 2a(iii)

Apply a two-sample t-test to determine whether there is sufficient evidence at the 5% level of significance, to support the claim that both classes are equally proficient at math.

(6 marks)

Question 2a(iv)

Construct a 95% CI on the difference in mean score that can be used to test the claim in Question 2(a)(iii).

(4 marks)

Question 2a(v)

In the question above, it was stated "Assume that the test scores from the two classes are approximately normal with same population variance". Do you think these assumptions are reasonable in this context?

(4 marks)

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Question 2b

Some Photoelectric smoke detectors detect smoke using a light emitter and a light receiver (Think of it as a laser pointer pointing a beam of light to a sensor). When smoke enters the smoke detector, it reduces the amount of light that reaches the sensor because smoke will scatter the light from the emitter.

The smoke detector will sound the alarm if the Light detected by the Sensor is reduced by P%.

Question 2b(i)

Explain TWO types of errors that could be made by smoke detectors and the possible costs/ impact of each type of error.

(7 marks)

Question 3a

The results of the 42.195km full marathon for the trials and actual run of the latest inter-university competition are summarised in Table Q3(a) below.

	Final Run			
Trial Run		**Under 4 hours**	**4 - 4.5 hours**	**More than 4.5 hours**
	Under 4 hours	58	50	46
	4 - 4.5 hours	62	115	130
	More than 4.5 hours	19	40	80

Table Q3(a)

We are interested to determine if the performance in one run is associated with the performance in the other run.

Question 3a(i)

Name an appropriate test procedure for analysing the data.

(1 mark)

Question 3a(ii)

Apply the statistical test at the 5% level of significance.

i. State the null and alternative hypotheses.
ii. Compute the test statistic.
iii. Compare the test statistic against either critical region or p-value.
iv. Draw conclusions in the context of the question.

(16 marks)

Question 3b

The water bills for all 5 Room HDB households in Hougang have an unknown distribution with mean \$160 and standard deviation \$28. A survey of 120 households were conducted.

Question 3b(i)

Compute the probability that the mean water bill will be between \$153 and \$166.

(3 marks)

Question 3b(ii)

What is the probability that the mean water bill will be within \$6 of the population mean?

(2 marks)

Question 3b(iii)

The Head Statistician wants to increase the sample size so that the width of the 95% confidence interval of the sample mean can be decreased to just \$2. Compute the minimum sample size to meet this criteria.

(3 marks)

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Question 4

A student has to answer 70 Multiple Choice Questions (MCQ) in the latest statistics test. For each question, there are 3 Choices, and only 1 answer is correct.

There was a particular student, Paul, who was completely unprepared for the test, and he answered the questions by randomly selecting one of the 3 Choices for each question.

Question 4a

The passing grade for the test is 40%, and assume that all questions are worth the same number of marks.

Apply the exact binomial model, compute the probability of Paul attaining a pass in the exam.

(5 marks)

Question 4b

Apply the normal approximation method, compute and estimate the probability of Paul getting a pass.

Verify the conditions for which the normal approximation to binomial will be appropriate, and comment on your answer.

(7 marks)

Question 4c

Now we assume another scenario where the 70 questions are NOT worth the same number of marks. The First 40 questions (Q1 to Q40) are worth 1 mark each, and the Last 30 questions (Q41 to Q70) are worth 2 marks each.

Question 4c(i)

Let the total score due to the First 40 Questions (Q1 to Q40) be T_1. Apply the normal approximation to find a suitable statistical model for T_1.

(3 marks)

Question 4c(ii)

Let the total score due to the Last 30 Questions (Q41 to Q70) be T_2. Apply the normal approximation to find a suitable statistical model for T_2.

(3 marks)

Question 4c(iii)

Using your answer to Question 4(c)(i) and 4(c)(ii), compute the probability for Paul to get a pass.

(5 marks)

Question 4c(iv)

Comment on your findings.

(2 marks)

----- END OF PAPER -----

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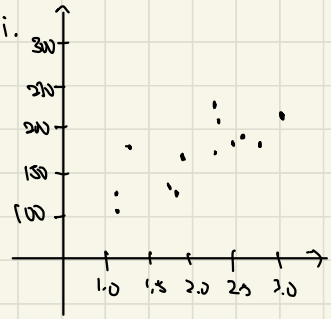
1(a) i.



There appears to be a moderate linear relationship between x and y.

ii. $n = 13$

$\sum x = 26.66$ $\quad$ $\sum x^2 = 58.8404$

$\sum y = 2265$ $\quad$ $\sum y^2 = 419423$

$\sum xy = 4911.8$

$\hat{y} = B_0 + B_1 x$

$B_0 = \bar{y} - B_1 \bar{x}$

$B_1 = \frac{S_{xy}}{S_{xx}} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \sum x \sum x}$

$\bar{x} = \frac{1}{n}\sum x = 2.050769$

$\bar{y} = \frac{1}{n}\sum y = 174.2308$

$B_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \sum x \sum x} = 64.03038$

$B_0 = \bar{y} - B_1 \bar{x} = 42.91924$

iii. $\hat{y}(x) = B_0 + B_1 x$

$\hat{y}(1.5) = 138.948$

iv. $r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - \sum x \sum x}\sqrt{n\sum y^2 - \sum y \sum y}} = 0.8301063$

1(b) i. let $L = 91.9631$

let $U = 96.0369$

$n = 75$

sample mean $\bar{x} = \frac{1}{2}(U+L) = 93.97$

$z_{critical} = \text{qnorm}(1-\alpha/2)$ where $\alpha = 0.05$

$= 1.959964$

$\bar{x} + z_{critical}\frac{s}{\sqrt{n}} = U$

$s = \frac{\sqrt{n}}{z_{critical}}(U - \bar{x})$

$= 9.132759$

sample variance $= s^2 = 83.40729$

ii. $H_0: \mu = 90g$

$H_1: \mu > 90g$

$T = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = 3.764103$

$T_{critical} = \text{qt}(0.90, n-1) = 1.293097$

since $T > T_{critical}$, reject H_0.

There is sufficient evidence the company has underestimated the mean mass of the pastries

iii. The manufacturing manager intends to test

$H_0: \mu = q$

$H_1: \mu \neq q$

at $\alpha = 0.01$,

$t_{critical} = \text{qt}(1-\alpha/2, n-1)$

$= 1.992543$

$\frac{\bar{x}-q}{\frac{s}{\sqrt{n}}} \leq t_{critical}$

$q \geq \bar{x} - t_{critical}\frac{s}{\sqrt{n}}$

$q \geq 91.88074$

min $q = 92$

2(a) i. $H_0: \mu_A = \mu_B$

$H_1: \mu_A \neq \mu_B$

ii. $\bar{x}_A = 80$ $\quad s_A = 3$ $\quad n_A = 11$

$\bar{x}_B = 78$ $\quad s_B = 4$ $\quad n_B = 14$

pooled variance $s_p^2 = \dfrac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$

$= 12.95652$

$s_p = 3.599517$

iii. $\alpha = 0.05$

$t_{critical} = qt(1 - \alpha/2, n_A + n_B - 2) = 2.068658$

$t = \dfrac{(\bar{x}_A - \bar{x}_B) - (\mu_A - \mu_B)}{s_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} = 1.379038$

since $|t| < t_{critical}$, don't reject H_0

There is sufficient evidence to conclude both classes are equally proficient at mathematics

iv. lower bound $= (\bar{x}_A - \bar{x}_B) - t_{critical}\, s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$

$= -1.000$

upper bound $= (\bar{x}_A - \bar{x}_B) + t_{critical}\, s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$

$= 5.000$

$\therefore$ confidence interval $= (-1.000, 5.000)$

v. Yes, the assumption is reasonable since the difference between the 2 standard deviation isn't too big, so it doesn't contradict the assumption

2(b) i. Type 1 error happens when the null hypothesis is rejected wrongly. This is also known as a false positive. In this case, the smoke detector sounds an alarm when there is no fire. Other than annoyance, there is no damage.

Type 2 error happens the null hypothesis was not rejected when it should be rejected, aka false negative. In this case, the the smoke detector did not sound the alarm when there is a fire. In this case, the fire could have led to damage and injuries.

3(a) i. Chi-square test for independence

ii. i. H_0: the attributes of the 2 categorical variables are independent

H_1: the attribute of the 2 categorical variables are not independant

ii. total frequency $N = 600$

frequency for $r_1 = 154$ frequency for $c_1 = 139$

" $r_2 = 307$ " $c_2 = 205$

" $r_3 = 139$ " $c_3 = 256$

observed: $O_{i,j}$

	$j=1$	$j=2$	$j=3$
$i=1$	58	50	46
$i=2$	62	115	130
$i=3$	19	40	80

expected: $E_{i,j} = \frac{r_i c_j}{N}$

	$j=1$	$j=2$	$j=3$
$i=1$	35.46	52.62	65.91
$i=2$	71.12	104.89	130.99
$i=3$	32.20	47.49	59.31

partial χ^2: $\frac{(O_{i,j}-E_{i,j})^2}{E_{i,j}}$

	$j=1$	$j=2$	$j=3$
$i=1$	13.9468	0.13013	5.9104
$i=2$	1.1699	0.97413	0.007852
$i=3$	5.4123	1.1818	9.2203

$\chi^2 = 35.97436$

$df = (\text{rows}-1)(\text{clm}-1)$

$= (3-1)(3-1)$

$= 4$

$\chi^2_{critical} = \text{qchisq}(0.95, 4)$

$= 9.487729$

Since $\chi^2 > \chi^2_{critical}$, reject H_0. There is sufficient evidence the results for trial & final runs are not independent

3(b) let X be the water bill for a 5-rm HDB Household

$X \sim N(160, 28^2)$

$\bar{X} \sim N(160, \frac{28^2}{n})$ where $n = 120$

i. $P(153 < \bar{X} < 166)$

$= P(\bar{X} < 166) - P(\bar{X} < 153)$

$= \text{pnorm}(166, 160, \frac{28}{\sqrt{n}}) - \text{pnorm}(153, 160, \frac{28}{\sqrt{n}})$

$= 0.9874621$

ii. $P(154 < \bar{X} < 166)$

$= P(\bar{X} < 166) - P(\bar{X} < 154)$

$= \text{pnorm}(166, 160, \frac{28}{\sqrt{n}}) - \text{pnorm}(154, 160, \frac{28}{\sqrt{n}})$

$= 0.9810542$

iii. $2 z_{\text{critical}} \frac{\sigma}{\sqrt{n}} < 2$

$z_{\text{critical}} \frac{\sigma}{\sqrt{n}} < 1$

$\frac{1}{\sqrt{n}} < \frac{1}{\sigma z_{\text{critical}}}$

$\sqrt{n} > \sigma z_{\text{critical}}$

$n > (\sigma z_{\text{critical}})^2$

$\min n = \lceil (\sigma \cdot z_{\text{critical}})^2 \rceil$

$= \lceil (\sigma \cdot \text{qnorm}(1 - \alpha/2))^2 \rceil$

$= \lceil 3011.704 \rceil$

$= 3012$

The min. sample size is 3012.

4(a) let X be no. of question Paul answered correctly

$X \sim Bin(70, \frac{1}{3})$

40% is pass $\Rightarrow$ Paul must answer at least 28 questions correctly.

$$P(\text{Paul passes}) = P(X \geq 28)$$
$$= 1 - P(X \leq 27)$$
$$= 1 - \text{pbinom}(27, 70, \tfrac{1}{3})$$
$$= 0.1456406$$

(b) $np = 23.33333$

$n(1-p) = 46.66667$

$\therefore$ The normal approximation is valid.

let Y be normal approximation of X

$E(Y) = np = 23.33333$

$var(Y) = np(1-p) = 15.55556$

$Y \sim N(23.33333, 15.55556)$.

$$P(\text{Paul passes}) = P(X \geq 28)$$
$$= 1 - P(X \leq 27)$$
$$= 1 - P(Y \leq 27.5)$$
$$= 1 - \text{pnorm}(27.5, 23.33333, \sqrt{15.55556})$$
$$= 0.14538$$

(c) i. $X_1 \sim Bin(40, 1/3)$

$np = 13.33333$

$n(1-p) = 26.66667$

$\therefore$ normal approximation of X_1 is valid

let Y_1 be the normal approximation of X_1

$E(Y_1) = np = 13.33333$

$var(Y_1) = np(1-p) = 8.88889$

$Y_1 \sim N(13.33333, 8.88889)$

$T_1 = Y_1$

$\therefore T_1 \sim N(13.33333, 8.88889)$

4(c) ii. $Y_2 \sim Bin(30, 1/3)$

$np = 10$

$n(1-p) = 20$

$\therefore$ normal approximation is valid

let $\tilde{Y}_2$ be normal approximation of Y_2

$E(Y_2) = np = 10$

$var(Y_2) = np(1-p) = 6.66667$

$\tilde{Y}_2 \sim N(10, 6.66667)$

$T_2 = 2Y_2$

$E(T_2) = 2E(Y_2) = 20$

$var(T_2) = 2^2 var(Y_2) = 26.66667$

$\therefore T_2 \sim N(20, 26.66667)$

iii. $T_1 \sim N(13.33333, 8.88889)$

$T_2 \sim N(20, 26.66667)$.

let $T = T_1 + T_2$

$\therefore T \sim N(33.33333, 35.55556)$

$\phi(\text{Paul passes}) = P(Y_1 + 2Y_2 \geq 40)$

$= P(T_1 + 2T_2 \geq 39.5)$

$= P(T \geq 39.5)$

$= 1 - P(T < 39.5)$

$= 1 - \text{pnorm}(39.5, 33.33333, \sqrt{35.55556})$

$= 0.1505255$